

only contains Sanskrit words but contains Sanskrit phrases also. I found out that it is not just a dictionary, but also helps you with Sanskrit to English translation and vice versa, which makes it a very useful tool. The dictionary even accepts Unicode based Sanskrit letters as input for searching the content. Moreover, it is an editable dictionary to which any registered user can contribute.

Sanskrit fonts and the open source initiative

There are a lot of open source word processing packages like OpenOffice Writer and LibreOffice Writer. But how many of these packages support Sanskrit fonts? Well, I came across many Sanskrit fonts which can be used in these word processing packages. Two such fonts that caught my attention were Lohit Devanagari and Kruti Dev. Lohit Devanagari is the default font. Kruti Dev is a more recently developed font. Both these fonts can be used with LibreOffice Writer and produce good quality Sanskrit text. Currently, around forty variants of the Kruti Dev font are available, including variants like bold, italics, thin, wide, etc. Figure 1 shows the famous mantra ‘Satyameva Jayate’ (truth alone triumphs) from the *Mundaka Upanishad* written in LibreOffice Writer with the Lohit Devanagari font.

LaTeX and Sanskrit

LaTeX is a word processor and a document markup language. Word processing packages like LibreOffice Writer use a page containing formatted text, whereas LaTeX uses a page containing plain text along with markup tags specifying the effects to be produced on the plain text. A LaTeX document has a .tex extension and on processing with a LaTeX engine will produce a PDF or DVI document as the output.

It is possible to typeset Sanskrit text using LaTeX. But standard keyboards do not have the Devanagari alphabet in them. So we need a transliteration scheme for translating text written in the English alphabet to Devanagari. Unlike usual LaTeX documents with an extension .tex, if your document contains the Devanagari script, then the extension is usually .dn. This file is then pre-processed with a macro to produce the corresponding .tex file. This .tex file is further processed by a LaTeX engine to produce the PDF or DVI output file.

So, unlike processing documents containing English letters, if you are processing Devanagari script, you need to use a two-step process. The C program called *devnag.c* to pre-process LaTeX files containing Devanagari script can be downloaded from the Comprehensive TeX Archive Network (CTAN) at www.ctan.org/tex-archive/language/devanagari/velthuis/. The file contains 2562 lines of excellent code — one of the best I have ever come across. Again, to my surprise, I found out that the program was written by Frans J. Velthuis, a Dutchman associated with the University of Groningen in Netherlands! You just need to compile the C program to produce an executable, and



Figure 1: Sanskrit in LibreOffice Writer

then with this executable you need to pre-process the .dn file to produce the .tex file. The code below illustrates the compilation and execution of the C program *devnag.c* for pre-processing *sanskrit.dn* to produce *sanskrit.tex*:

```
[root@deepu Sanskrit]# gcc devnag.c -o devnag
[root@deepu Sanskrit]# ./devnag sanskrit.dn
```

The transliteration scheme for Devanagari is clearly explained in the user manual available with the C file *devnag.c*. But obtaining just a .tex file is not our aim. We need to process the file *sanskrit.tex* further to obtain the PDF or DVI output file. For this you need to install a package called *Devanagari*. The style sheet for this package, called *Devanagari.sty*, is also available along with the other resources in the CTAN link mentioned earlier. The code below shows the file *sanskrit.dn* before it's processed with the pre-processor program. If you observe the code carefully, you will see that whatever text comes inside the \dn environment is treated as Sanskrit by default.

```
\documentclass[12pt]{article}
\usepackage{devanagari}
\begin{document}
{\dn
\begin{center}
\Huge\framebox[6cm]{.o gurave nama.h}
\end{center}
}
\end{document}
```

The pre-processor program will take care of the transliteration scheme of the Devanagari characters. After processing, the file *sanskrit.dn* will yield a file called *sanskrit.tex*. The code below shows the file *sanskrit.tex*:

```
\def\DevnagVersion{2.15}\documentclass[12pt]{article}
\usepackage{devanagari}
\begin{document}
{\dn
\begin{center}
\Huge\framebox[6cm]{: \7{g}rv? nm,}
\end{center}
}
\end{document}
```

The file *sanskrit.tex* is further processed by the LaTeX engine to produce the PDF or DVI output file. Figure 2 shows the PDF file *sanskrit.pdf* produced by the file *sanskrit.tex* and